Predicting H5N1 Outbreaks Using Social, Search, & Environmental Data

Sponsor:

Dr. Bouchra Nasri

Co-Sponsors:

Muthu Venkataraman, Jason Lin

Team Members:

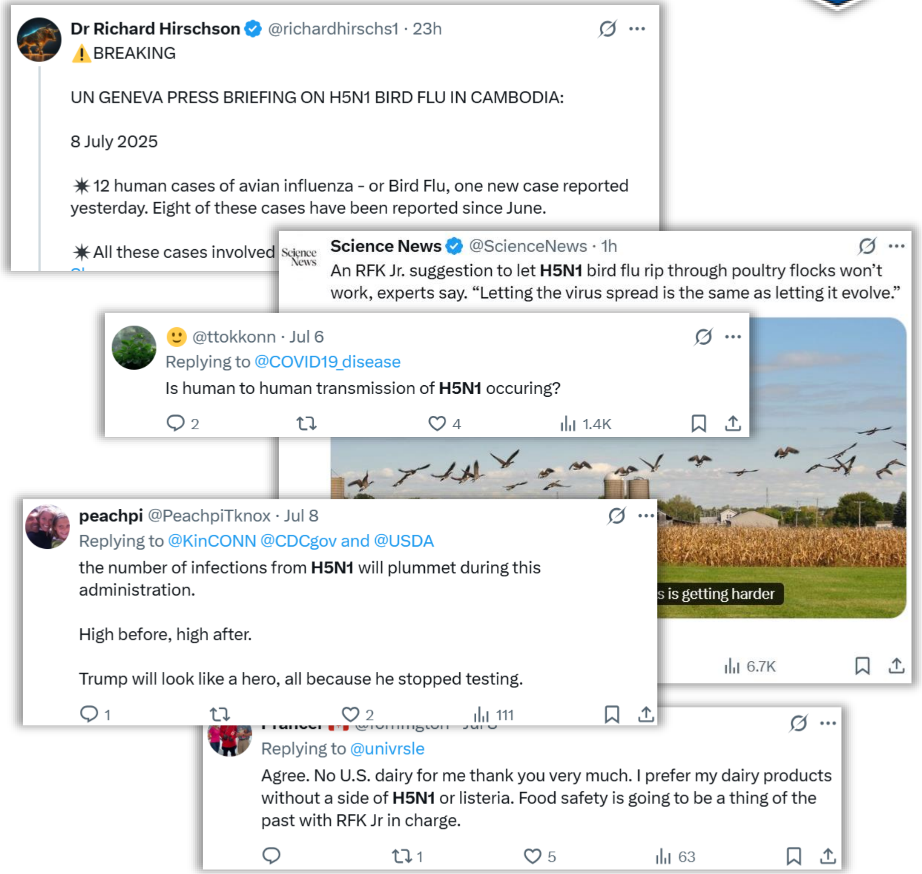
Byeongsu Kim, Gazi Saief Mahmud, Noura Almansoori, Renee Van Siclen, Spencer Beaupre

Project Overview

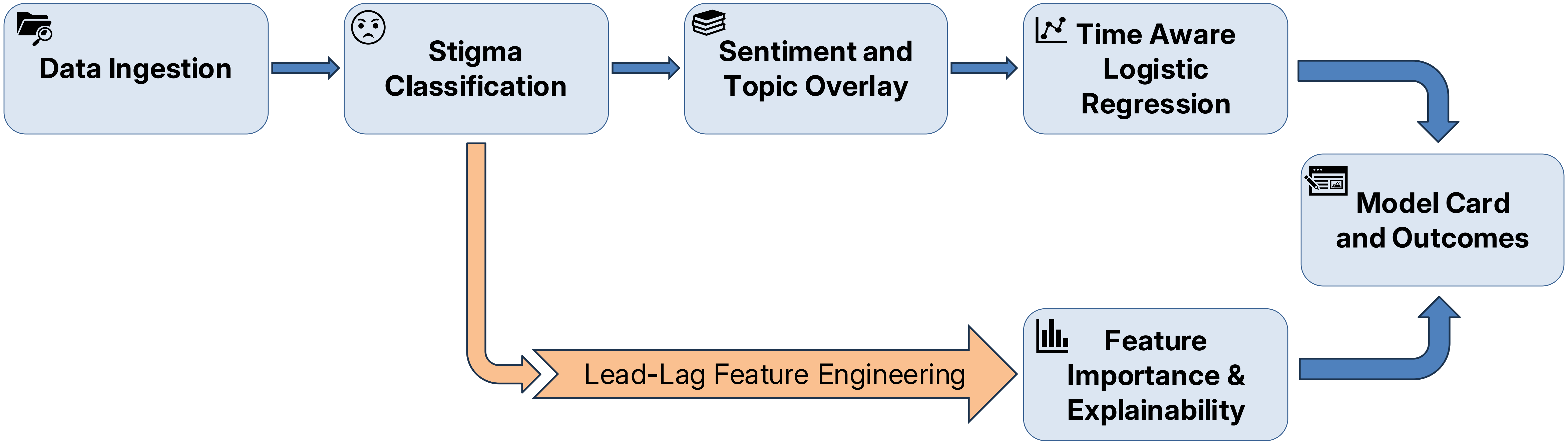
Our project explores how digital discourse affects public sentiment and risk perception around H5N1

To achieve this, we integrated the following data sources:

- **META** – digital discourse
- **WHO Disease Outbreak News (DON)** – human cases
- **FAO EMPRES-i** – confirmed animal outbreaks
- **GDELT** – near–real-time global media signals



End-to-End Modeling Pipeline



Stigma Labeling Methodology

Approach

Stigma categories were derived using a rule-based linguistic classification pipeline applied to cleaned text data

Label construction

Keyword lists were developed for four stigma-related constructs:

- **Fear**: e.g., fear, terrified, outbreak, deadly, threat, transmission
- **Blame** (merged into fear): e.g., failure, corrupt, negligence
- **Urgency**: e.g., urgent, breaking, alert, immediate, emergency
- **Humor/Dismissiveness**: e.g., lol, haha, meme, sarcastic

Text normalization & token extraction

A **hierarchical classification** procedure was applied: as follows

- Humor detection (highest precision)
- Fear detection
- Blame attribution (merged into fear)
- Urgency detection

Fallback rules

- Posts with strong negative valence were labeled *fear*
- Posts not satisfying any condition were labeled *neutral*

Examples: One-Hot-Encoded Stigma Labels & Sentiment Polarity

post_text	stigma_fear	stigma_urgency	stigma_neutral	sentiment_polarity
scientists at a scripps research symposium warn of h5n1 avian influenzas pandemic potential, urging urgent preparedness amid concerns over genetic mutations, inadequate infrastructure, and political hurdles.\n\n	FALSE	TRUE	FALSE	0.05
what would happen if there were no in the : medium\n\n could push heatwaves from us to europe : new sci\n\nthe industry must act faster to keep from starting a human : sci am\n\ncheck our latest \n	FALSE	TRUE	FALSE	-0.08
experts warn h5n1 bird flu could spark a pandemic\n\nhuman catch the flu through contact with infected poultry \n\ncan bird flu mutate to spread through the air?\n\nshivan chanana brings you this report\n\nwatch more:	TRUE	FALSE	FALSE	-0.02
in an updated report on 70 laboratory-confirmed highly pathogenic avian h5n1 cases in the us between march 2024-april 2025, the absence of human to human transmission to date is confirmed, with no known mutations of resistance to neuroaminidase inhibitors.	TRUE	FALSE	FALSE	-0.05
an 11-year-old boy from gurugram was reported as the country's first documented case and death of (h5n1) in humans.\n\nthe team of doctors and nurses, who treated the patient, is being monitored since july 16 for any influenza like illness.	TRUE	FALSE	FALSE	-0.09
the world health organization says that the child, australia's first case of h5n1 in a person, had traveled to kolkata from february 12 to february 19 and returned to australia on march 1	FALSE	FALSE	TRUE	0.06
the state of the bird flu virus, or h5n1, has evolved significantly in the past few months.\n	TRUE	FALSE	FALSE	-0.025

Lead-Lag Feature Engineering

Approach

We explicitly model when signals occur and whether they persist, rather than treating social signals as static snapshots

Lagged Features - Temporal Precedence

For each country–day signal (e.g., stigma fear, urgency, topic activation, sentiment polarity), values are shifted backward in time by fixed lags (1, 3, 7 days)

Key questions:

- Do fear or urgency increase prior to outbreak confirmation?
- Do specific topics (human cases, livestock infections) emerge earlier?
- Does sentiment polarity shift before official reporting?

Rolling Averages - Persistence & Momentum

- Encodes sustained concern rather than short-lived spikes.
- Signals are smoothed using rolling windows (e.g., 7 days) to capture emotional and narrative buildup over time.

Key questions:

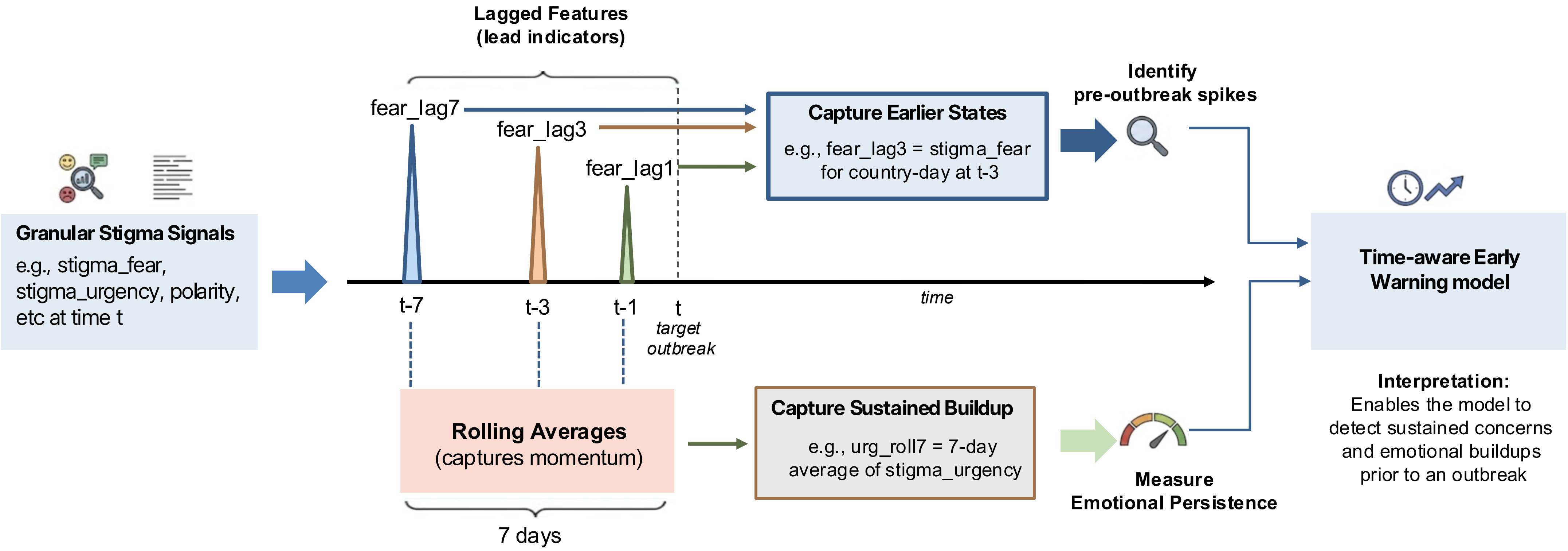
- Is fear sustained over multiple days?
- Does urgency persist rather than fluctuate?
- Is sentiment polarity stable and directional?

This lead–lag framework is applied consistently across Stigma dimensions , Topic Activations, and Sentiment Polarity

Lead-Lag Feature Engineering - Explained

Approach

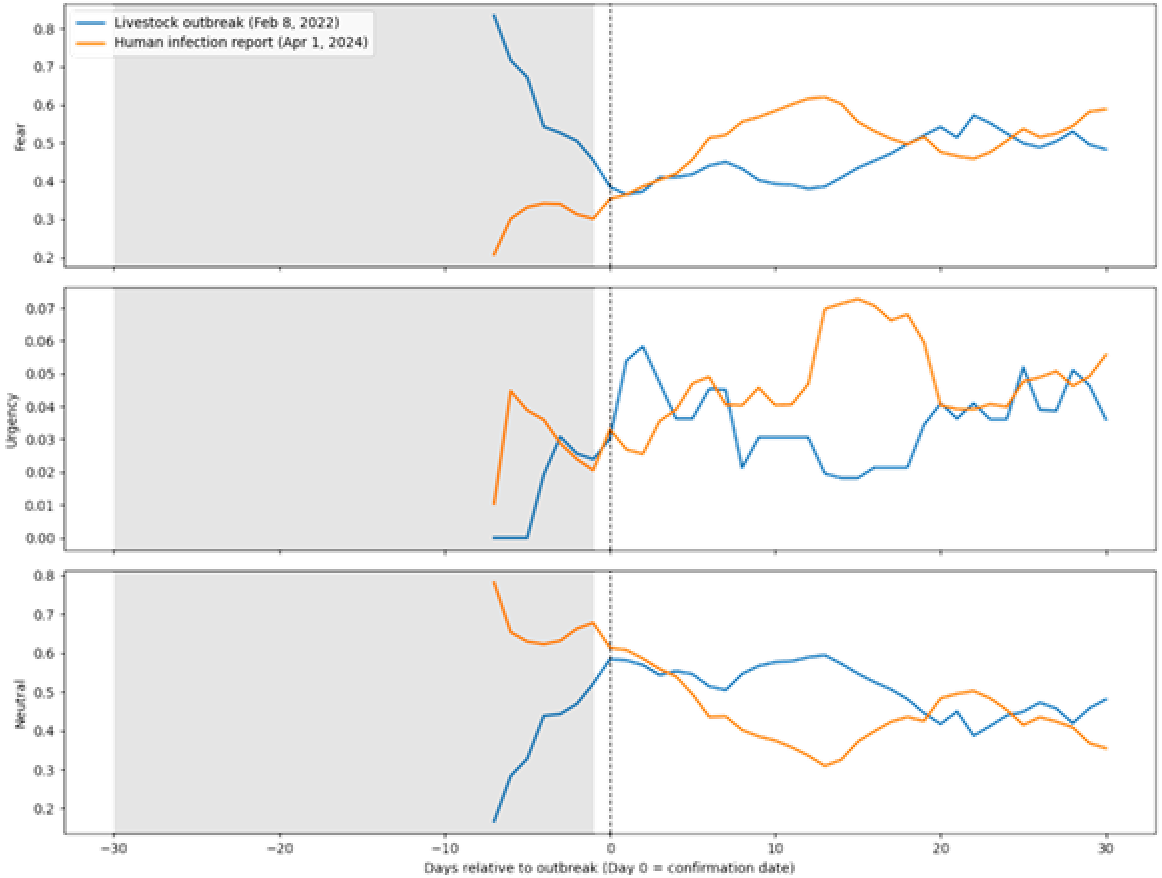
We explicitly model when signals occur and whether they persist, rather than treating social signals as static snapshots



Lead-Lag Feature Engineering - Example

The figure demonstrates that sustained increases in fear coupled with declining neutrality precede outbreak confirmation, while urgency peaks closer to or after confirmation.

- **Fear:** Fear-related discourse shows a clear lead–lag pattern for both outbreak types.
- **Urgency:** Urgency exhibits sharper, event-proximal responses for both event types.
- **Neutral:** displays an **inverse relationship** to fear and urgency.



Early Warning Model Construction

Model objective

- A **daily, country-level classifier** predicting whether a date falls within a pre-outbreak risk window
- Designed to **capture lead–lag relationships** between digital discourse and outbreak confirmations

Temporal signal design

- **Lagged features** were generated by shifting each stigma signal backward in time by 1, 3, and 7 days
- **Rolling averages** were computed over 7-day windows

Modeling choice

Class-weighted logistic regression

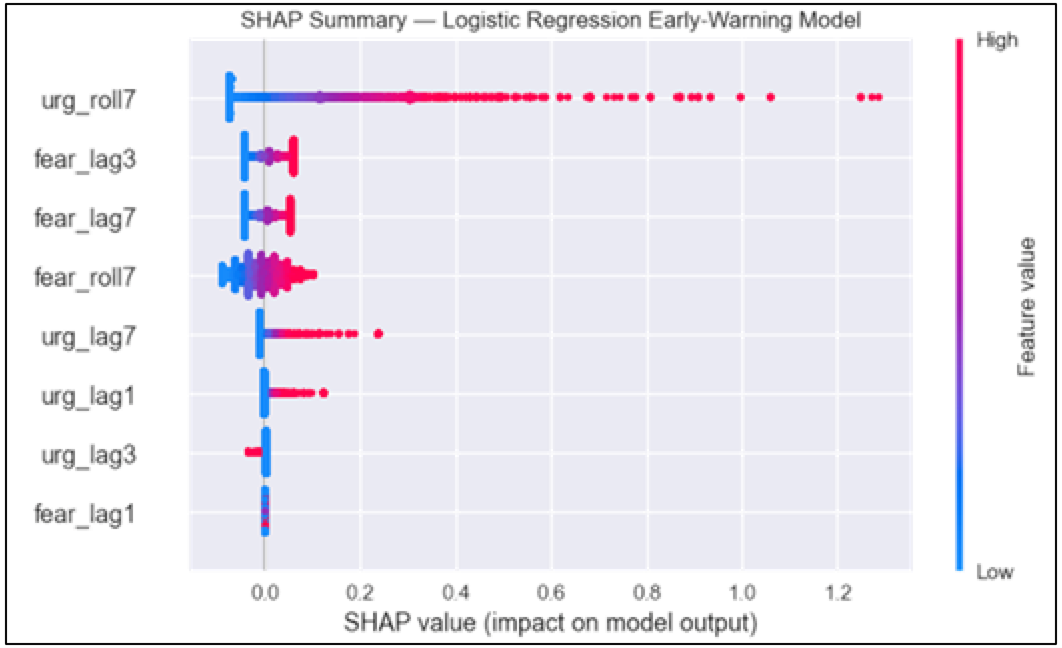
Evaluation metrics

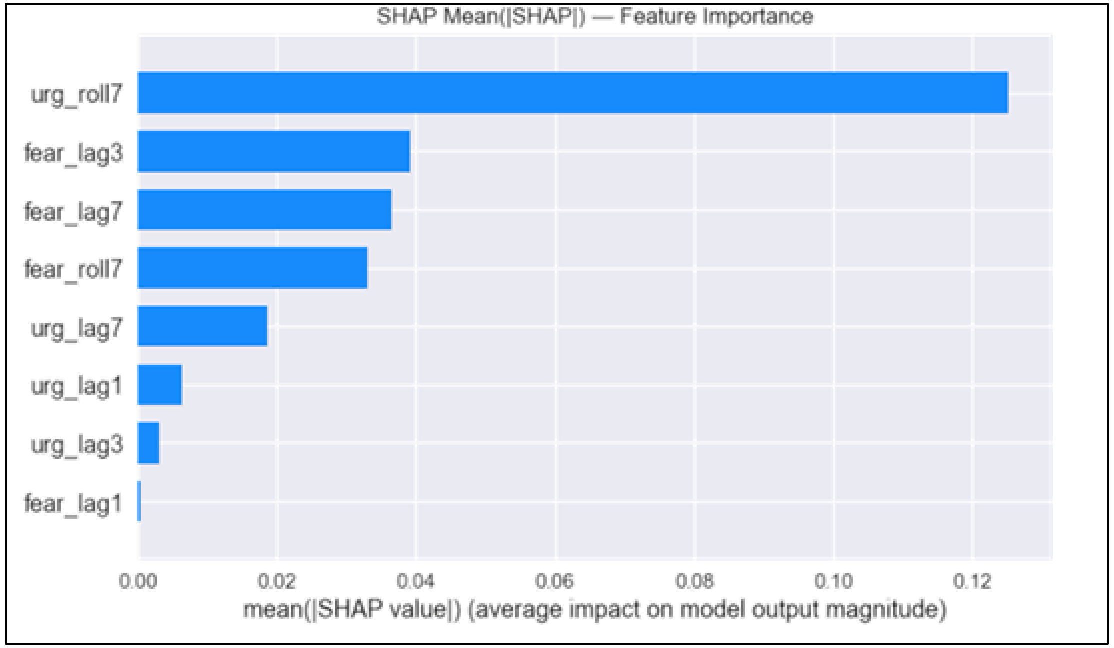
ROC-AUC, PR-AUC, recall, threshold sensitivity

Model Outcome (1/2)

Key findings

- Urgency is the dominant early-warning signal
- Fear signals with short and medium lags contribute meaningfully
- Short-term urgency spikes also influence predictions





Model Outcome (2/2)

Strengths

- Demonstrates meaningful early-warning signal (AUC = 0.73)
- High recall for early-warning periods (0.67–0.77)
- Stable performance under threshold tuning
- Transparent, interpretable structure

Limitations

- Low precision for early-warning periods (0.03–0.04)
- F1-score remains very low for outbreak class (0.06–0.07)
- Performance varies between countries
- Lag window method captures temporal signals but not complex dynamics

Implications

Early risk detection

Emotional trends online can signal emerging outbreaks before traditional surveillance

Critical communication window

Surges in fear or urgency indicate when public communication should be intensified

Future system design

Sustained emotional signals provide stronger predictive value than short-term spikes

Targeted messaging

Understanding how specific narratives couple with stigma allows for targeted communication

Study Limitations

- **Delayed access to Meta Content Library**, limiting time for extended exploratory analysis, feature refinement, and sensitivity testing
- Analysis relies on **English-language data and platform-specific user populations**, limiting cultural generalizability
- Rule-based **classification cannot fully capture nuanced expressions** such as sarcasm, irony, or meme-driven discourse
- **Variability in reporting timelines** across surveillance sources introduces noise in linking digital signals to outbreak events

Future Work

Continue to fine tune Advanced Forecasting Models:

- Improve Temporal Fusion Transformers (TFT) modeling
- CatBoost with time-aware cross-validation for tabular lead-lag features
- Explore reinforcement learning or agent-based models for behavioral amplification

Operationalize an Early Warning Dashboard:

- Real-time ingestion of stigma signals
- Country-level risk scores
- Visual overlays

Discussion